# Explainable AI for Academic Integrity Verification

Catching AI-written essays fairly, and checking if the student actually understands

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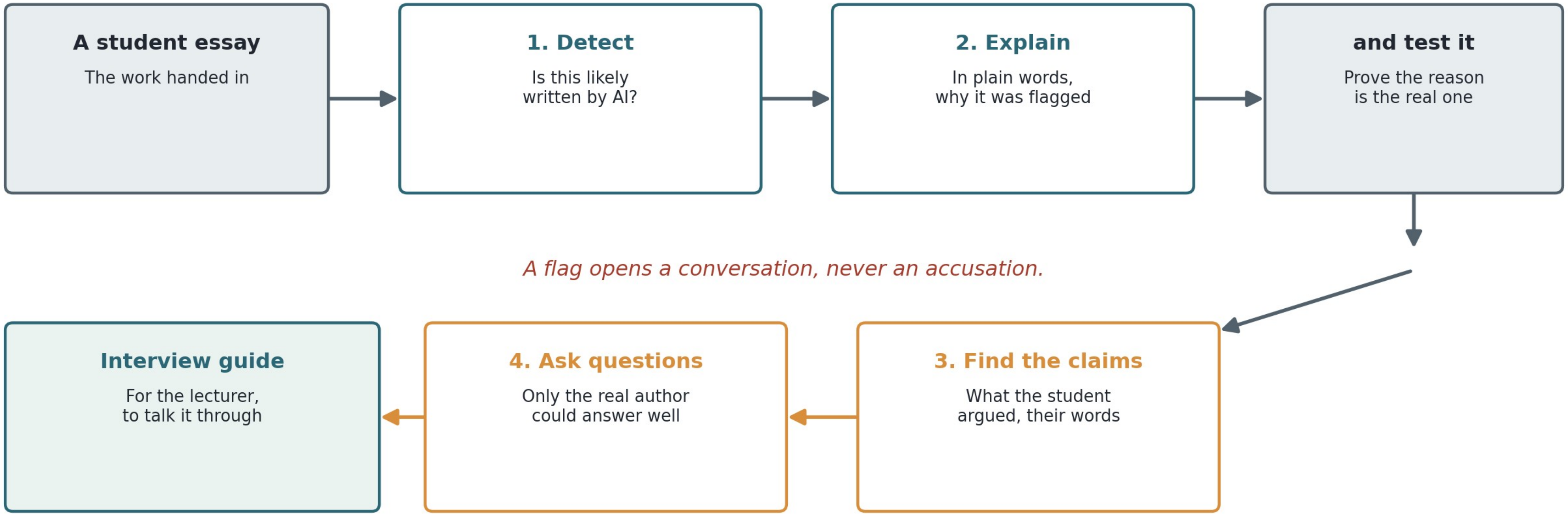
# The problem, in plain terms

- Students can now get an essay written by AI in seconds. Universities want to know when that has happened.
- The popular tools just print a number, like '82% AI'. A lecturer cannot defend that number if a student objects, and cannot tell if it is even right.
- Worse, those tools are unfair: they wrongly accuse students who write in a second language far more often. One Stanford study found 61% of those students' essays falsely flagged as AI, versus 5% for native writers (Liang et al., 2023).
- And a number answers the wrong question. It guesses HOW the essay was made, not whether the STUDENT understands what they handed in.

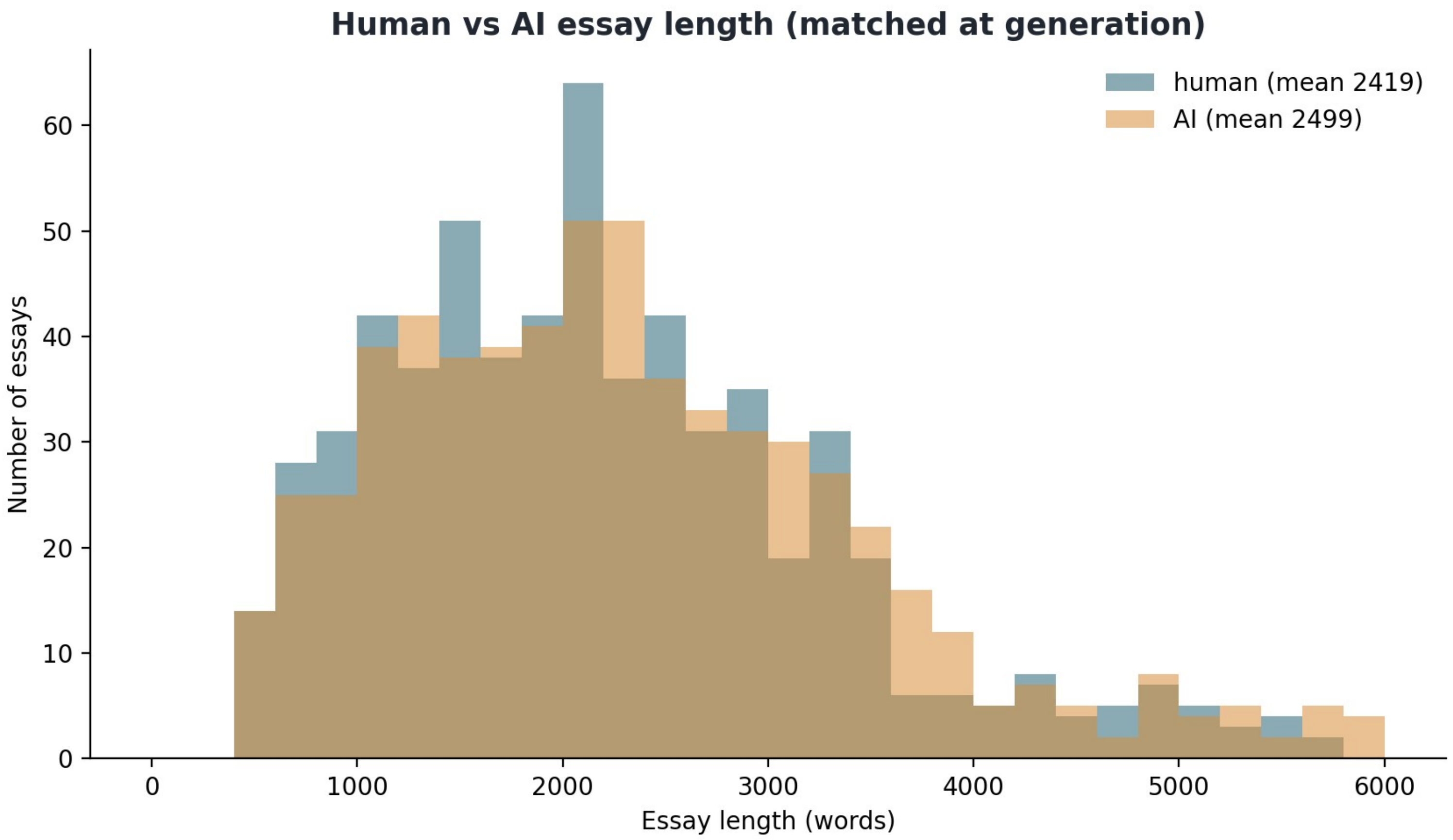
# My idea, in one picture

Do not just accuse. (1) Detect if it is likely AI. (2) Explain the flag in plain words, and prove the reason is real. (3) Find the student's own claims (the sentences where they make an argument). (4) Write questions only the true author could answer. Then hand the lecturer an interview guide. A flag starts a conversation, it does not end one.



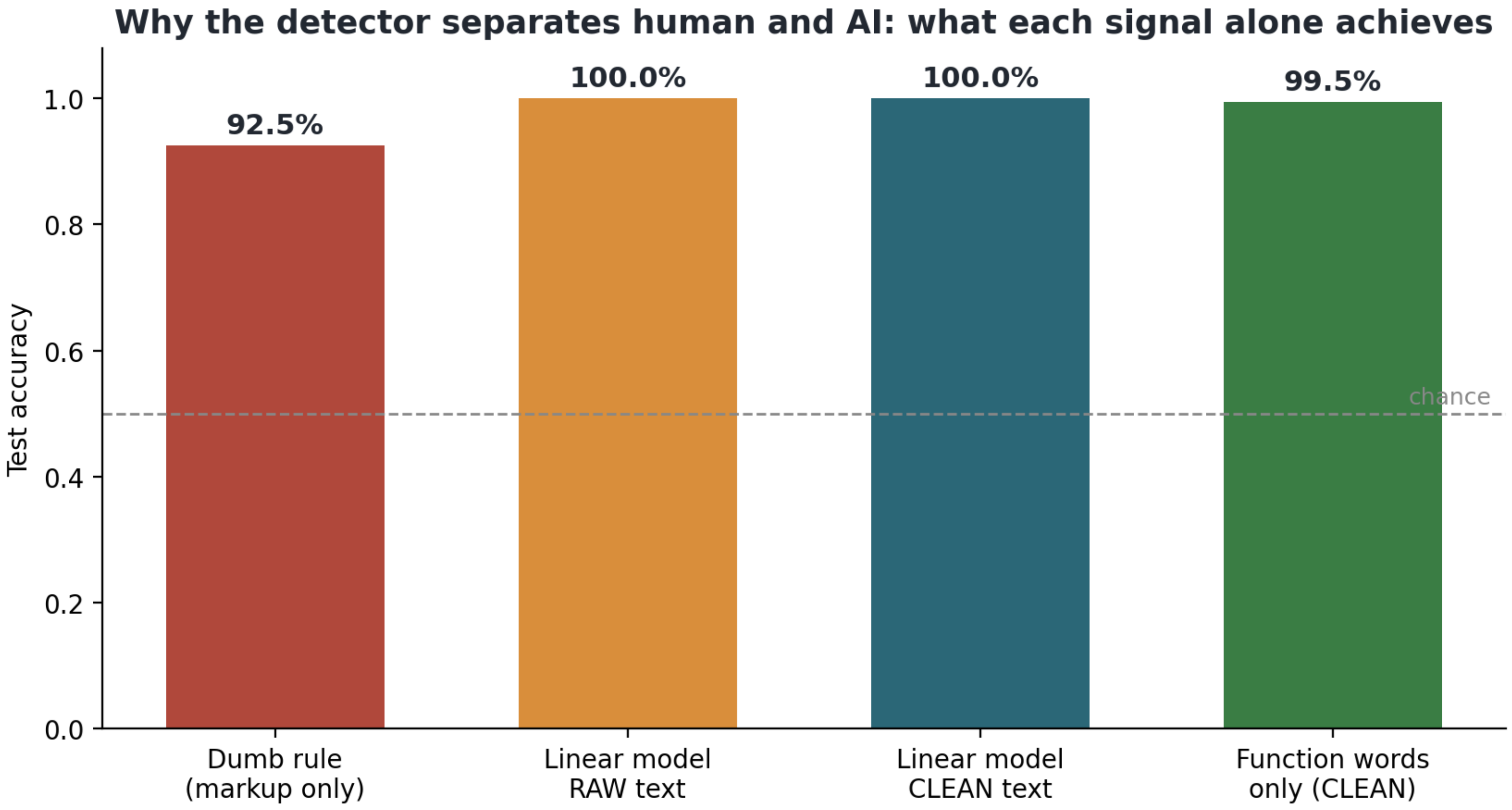
# Building block 1: a fair test set

640 real student essays (from the BAWE corpus: Alsop and Nesi, 2009) paired with 640 AI essays on the same topics at the same lengths (1,280 in total). The two length curves sit on top of each other, so the detector cannot cheat by simply noticing that AI essays are longer or shorter. It has to learn writing style.



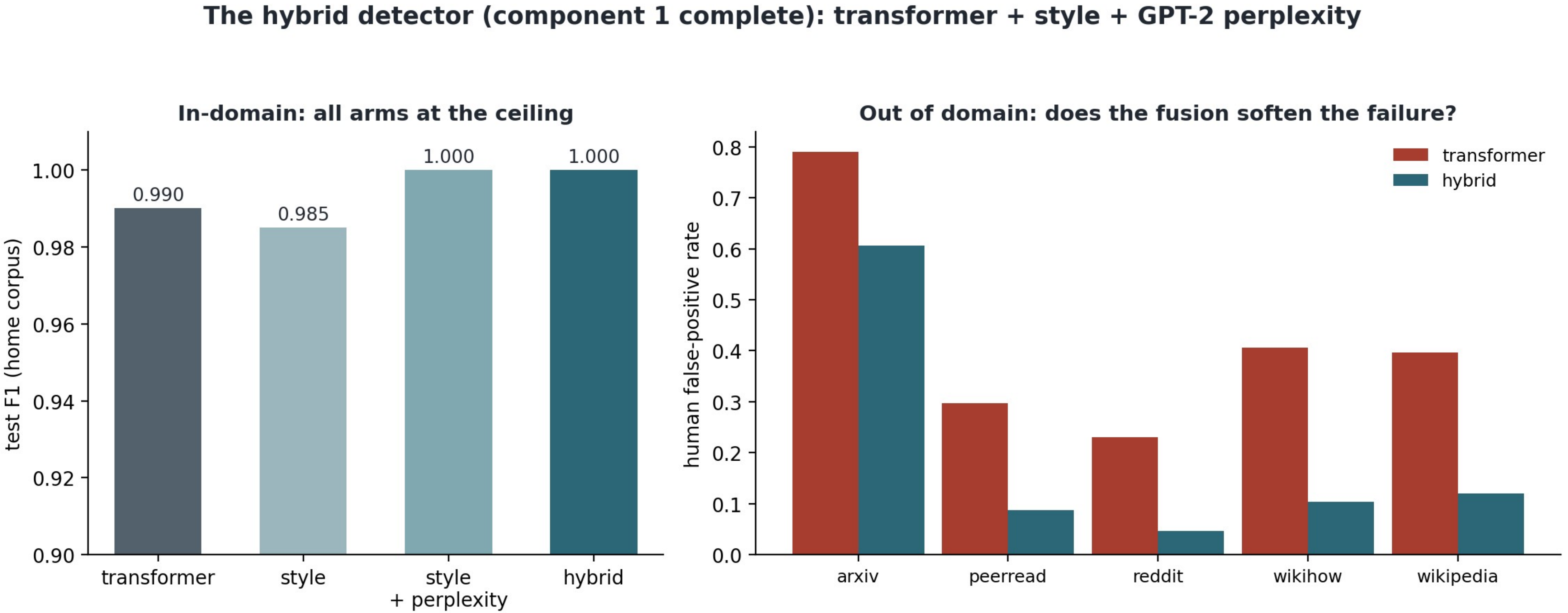
# Building block 1: I did not trust a perfect score

My first detector scored a perfect 100%. That is a warning, not a triumph. It turned out the real essays carried tiny hidden formatting tags (invisible leftovers from how the files were saved) the AI ones lacked, nothing to do with writing. I removed it and retrained; the honest score settled at about 99% on essays like the ones it learned from, now genuinely reading writing style.



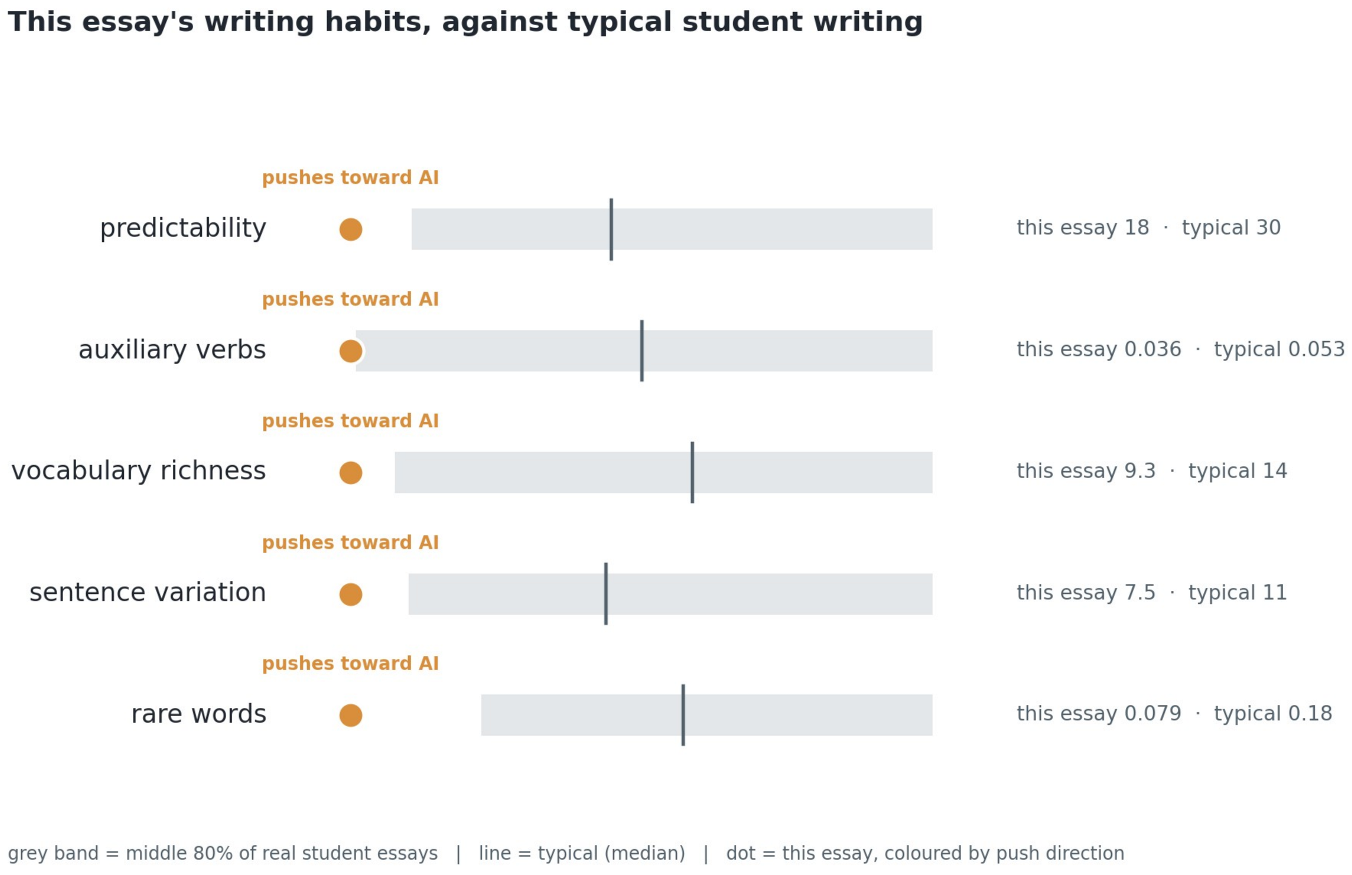
# Building block 1: made fairer, on purpose

The finished detector combines two ways of reading an essay: the exact words and phrases chosen, and the shape of the writing (sentence length, how varied it is), like checking both what someone says and how they say it. On its home essays both are near-perfect (left). It also caught 100% of test essays written by two commercial AI models it never saw in training, with zero false accusations of the humans. The hardest test is human writing from other worlds, like forum posts and short science summaries (right): the plain version wrongly flags up to 79% of them as AI, and the combined version cuts those false accusations three to eight times (science abstracts 79% down to about 61%). Fewer wrong accusations, the failure that matters most.



# Building block 2: explaining the flag (and where it is still weak)

Instead of a bare number, the lecturer gets this card, straight from the guide for a real flagged essay. Each row is one measurable writing habit. The grey band is the middle 80% of real student essays; the dot is this essay. Every dot sits outside the band: this essay is unusual in five separate ways, and each one is named in plain words. I also test that these reasons are real, not a plausible story. Honest gap: the card has not yet been tested on actual lecturers, and that test is a main job for the second half.



# Building blocks 3 and 4: from a flag to real questions

For a flagged essay the system finds the student's own claims and, for each, writes questions the true author could answer but a copier could not. Every question traces back to the student's exact sentences, so nothing is invented. Each is tagged by how much thinking it needs (Bloom's taxonomy: Anderson and Krathwohl, 2001).

**A claim the system found in the essay: the three set texts belong to different genres but share common themes, and the student argues this shows something about how literature works.**

Q1. How did you decide which themes the three texts actually share, and what in each text convinced you?

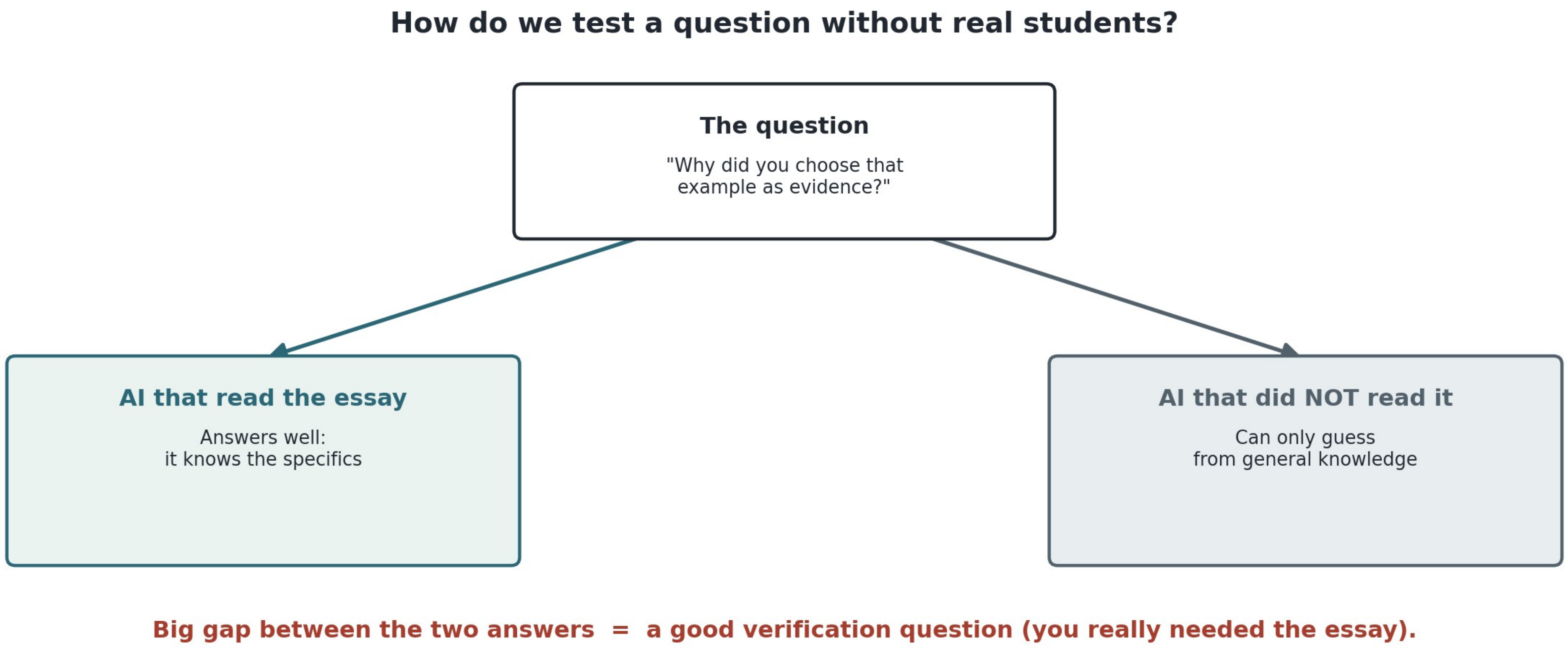
Thinking level: understand

Q2. How does that shared-theme point connect to the bigger argument of your essay?

Thinking level: analyse

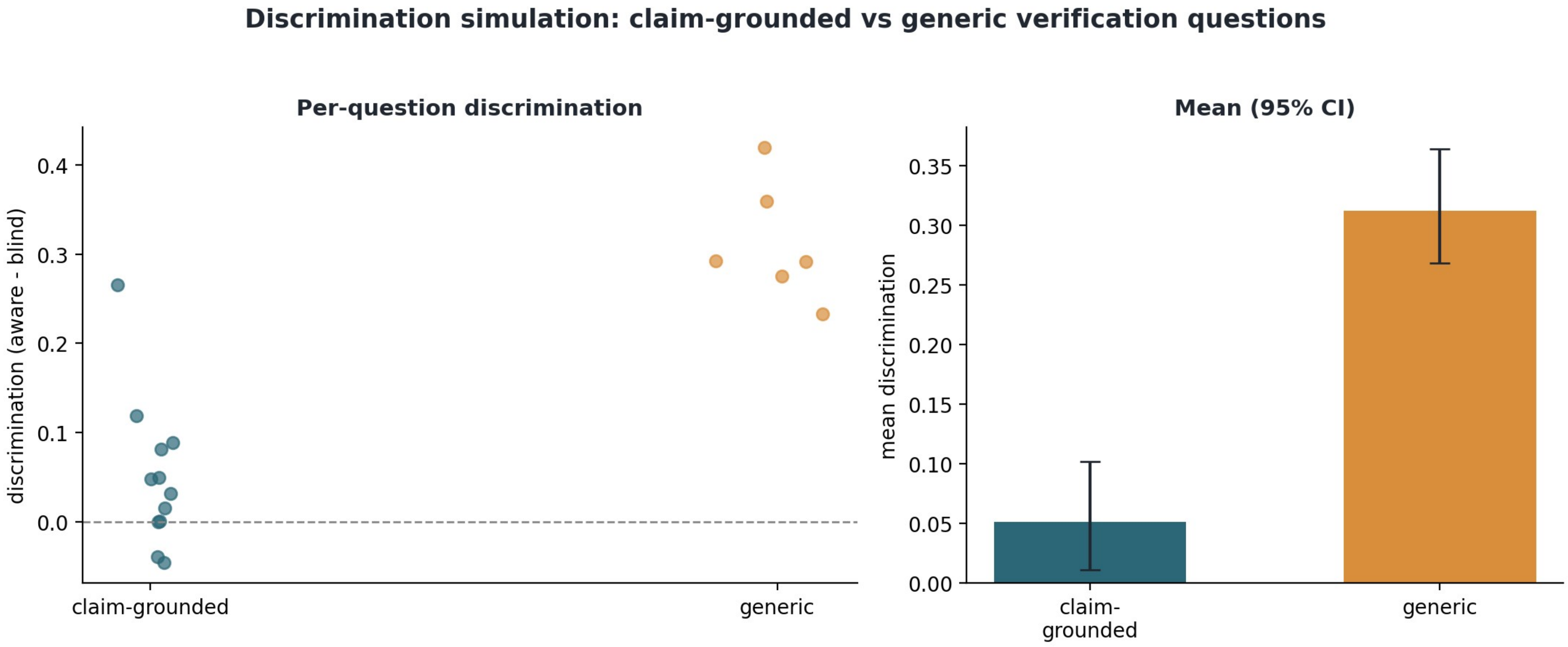
# How do you grade a question with no students yet?

A clever trick to test a question without a classroom. Show the question to two AIs: one that has read the essay and one that has not. If the one who read it answers much better, the question genuinely needs the essay, so it is a good verification question. A small gap means anyone could answer, so it is a weak one.



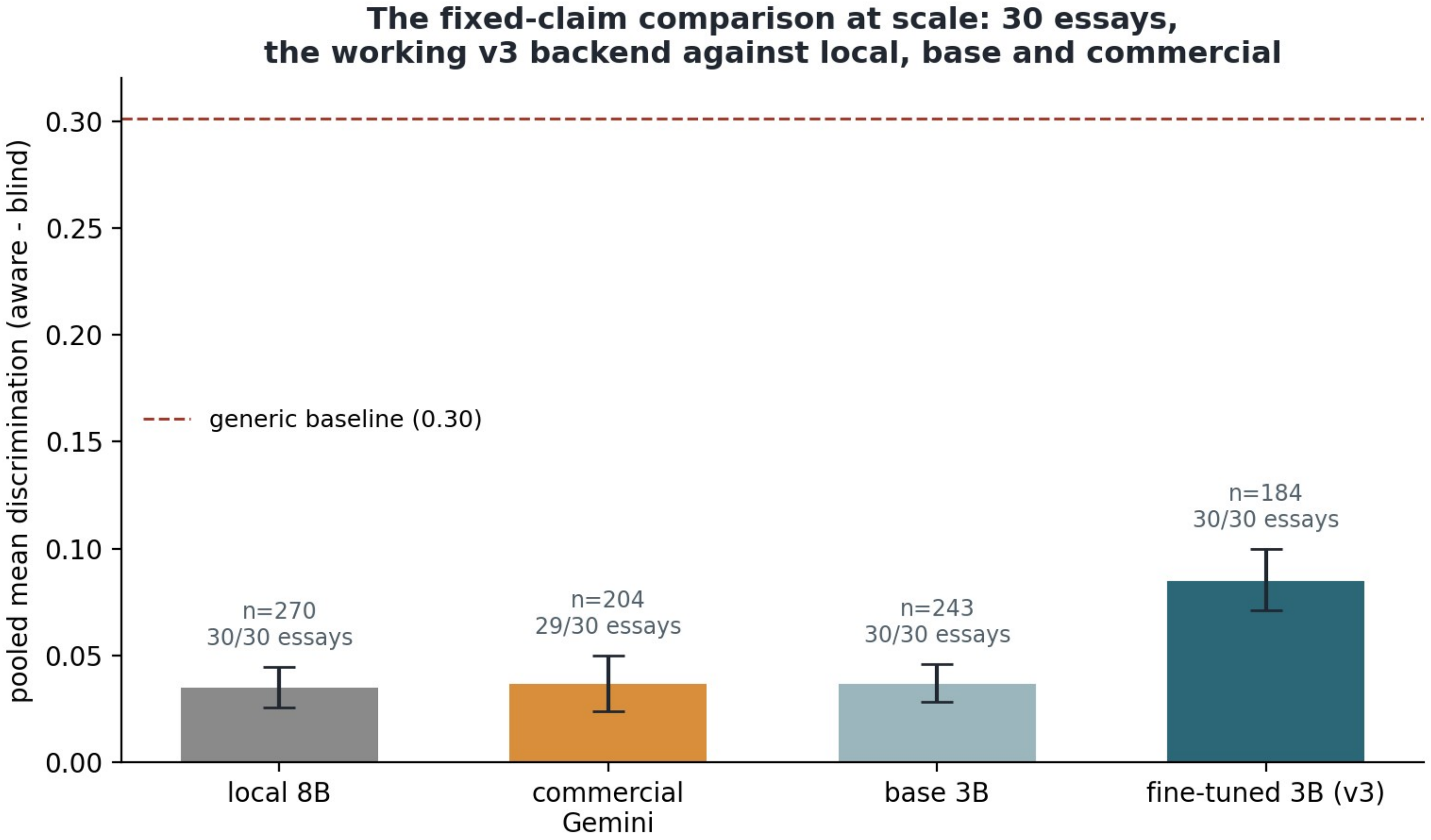
# A surprise that changed how I write questions

The trick immediately surprised me. Specific factual questions ('name the feature that...') scored low, because a knowledgeable AI answers them without the essay. Broad questions that force the student to walk through their OWN argument scored far higher. So verification questions should ask for the student's reasoning, not facts anyone could recite.



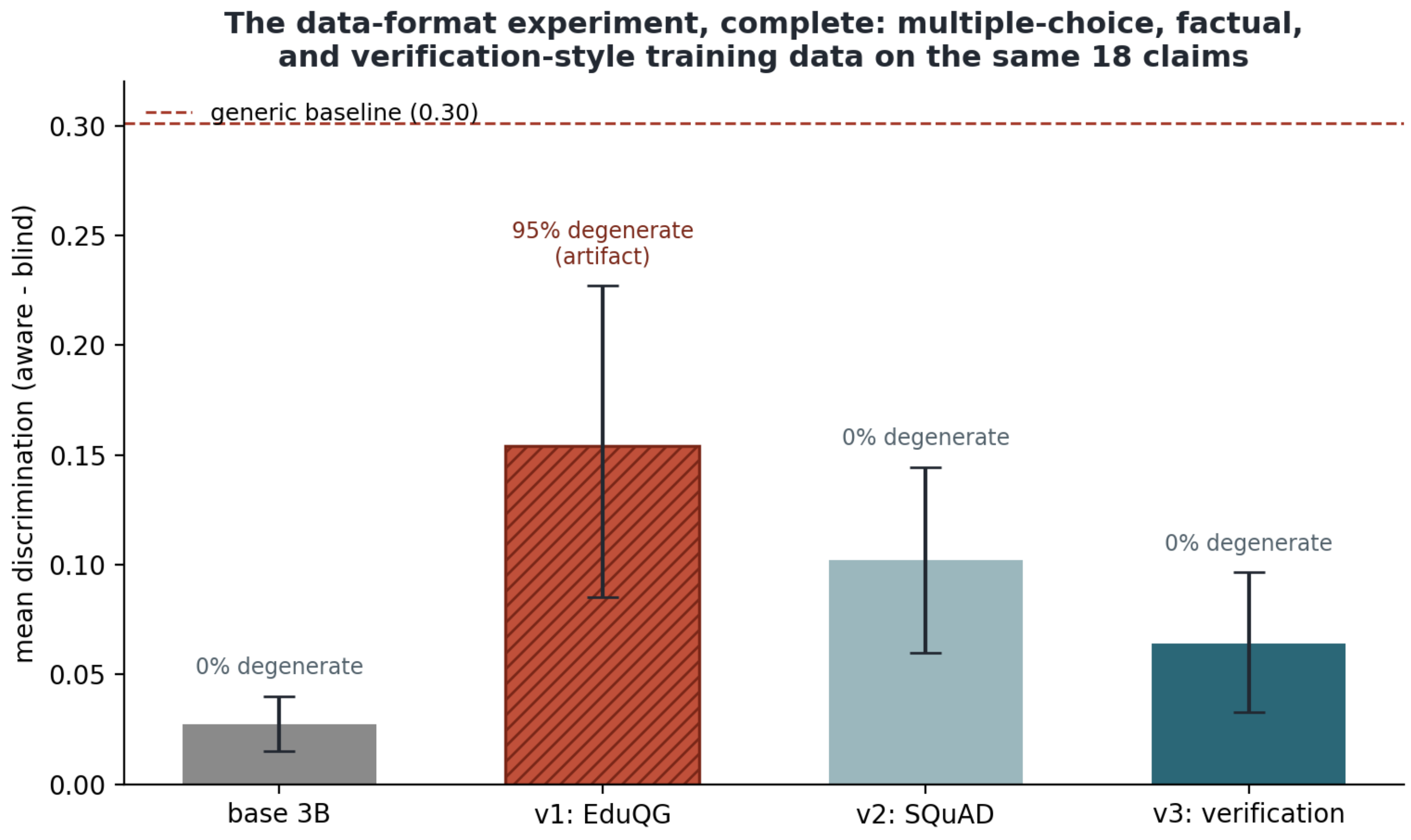
# Laptop model vs expensive cloud model: a fair race

Can a model on my own laptop match a paid cloud model? Normally each model picks which of the student's claims to ask about, and then the cloud model looked a bit better. But when every model must write about exactly the SAME claims (here across 30 essays and 901 questions), the plain models tie, and my fine-tuned laptop model wins: better questions than the cloud model on 24 of the 29 essays both covered, a gap far too consistent to be luck. The earlier cloud lead was about picking easier claims, not writing better questions.



# The result that lied to me (my favourite finding)

A model learns by copying thousands of examples. I trained my laptop model on the wrong examples, so it learned wrong: 95% of its questions (about 40 in the test) were broken, empty shells like 'Which of the following is correct?' with no options. And empty questions trick the two-AI test: with nothing real to answer, both AIs just guess, and by pure chance the one that read the essay lands a bit closer, so the gap looks big for the wrong reason. Its high score (tall red bar) was measuring emptiness. I proved the cause was the practice material, fixed it twice, and even so the raw score still ranks a shallow version above the correct one, so I never let the number decide alone. The ending came a week later: at thirty essays, the corrected model beats the cloud model fairly, with every question well-formed, and that claim I can finally stand behind because this time the quality and the number agree.



# Where the project is weak right now (and how I will fix it)

- Explainability is not clear enough. It convinces me, not yet a non-technical lecturer. Fix: plain visuals and plain words, tested on real readers. This is the biggest job.
- The questions have never met a real student. My two-AI trick is a stand-in. Fix: a small human study (the guide is ready to be that test).
- The training data is narrow: one source of human essays, one AI model. I have already added test essays from two more AI models (all caught). Fix: widen the training side the same way.
- Every question writer, mine included, still scores below a simple generic-question baseline on my quality measure, and the detector loses accuracy on kinds of writing it was not trained on. Fix: better question training data, and safeguards on unfamiliar writing.
- Samples started small. The main comparison now runs at 30 essays and 901 questions, and the judge check at 60 questions. Fix: keep scaling now that everything runs automatically.

# So where am I? About half way

Every building block now exists and the whole pipeline runs end to end, which was the hard part. But the second half is real work, not polishing: making the explanations genuinely understandable, testing on real answers, widening the data, strengthening the weak spots, and the final write-up. I am at the midpoint, with the more visible, human-facing half still ahead.

**we are here**

**BUILT SO FAR (June to mid-July)** | **SECOND HALF (mid-July to end of August)**

### Done

- ✓ Matched essay dataset (1,280)
- ✓ Hybrid AI-text detector
- ✓ Explainability + honesty test
- ✓ Claim extractor + Bloom classifier
- ✓ Question generation, both backends
- ✓ The evaluation and the fine-tune story
- ✓ The lecturer's guide, end to end

### Still to do

- Make the explanations clear and visual
- Validate with real student answers
- Widen the data (more AI models, domains)
- Push the local model further
- Fix the weak spots (see next slide)
- Discussion, polish, submit

# The plan for the rest of the summer, and the writing so far

- Now to end July: make the explanations clear and visual; integrate the trained components into the guide (done this week); scale the comparisons; make the detector more careful on unfamiliar kinds of writing.
- August: a small validation of the questions on real answers; widen the data with more AI models; final experiments.
- Late August: protected writing and revision. Hard rule: if time runs short, I cut analysis depth, never the write-up.
- Writing is kept in step with the work: an 88-page draft already exists, twelve chapters including Discussion and Conclusions, with 33 references each checked against the real paper (essential for a project about fake text).
- No human-participant ethics issue for the core work; all data is public and licensed; the output is always evidence for a conversation, never an accusation.

# One sentence to take away

- Do not accuse students with a number no one can defend. Detect fairly, explain honestly, and turn a flag into a fair conversation built from the student's own work.
- The method that runs through everything: never trust a score you cannot explain, and never trust one you have not tried to break. It caught three of my own results.
- Status: every piece built and running (detection at 99% with zero false accusations on unseen AI models, laptop question-writer beating the cloud model on a fair test), at the halfway point with the human-facing half still to do.

# References and tools

Liang, W., Yuksekgonul, M., Mao, Y., Wu, E. and Zou, J. (2023). GPT Detectors Are Biased Against Non-Native English Writers. Patterns, 4(7), 100779.

Alsop, S. and Nesi, H. (2009). Issues in the Development of the British Academic Written English (BAWE) Corpus. Corpora, 4(1), pp. 71 to 83.

Anderson, L. W. and Krathwohl, D. R. (2001). A Taxonomy for Learning, Teaching, and Assessing: A Revision of Bloom's Taxonomy of Educational Objectives. Longman.

He, P., Liu, X., Gao, J. and Chen, W. (2021). DeBERTa: Decoding-enhanced BERT with Disentangled Attention. ICLR 2021.

Lundberg, S. M. and Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS 2017.

DeYoung, J., Jain, S., Rajani, N. F., Lehman, E., Xiong, C., Socher, R. and Wallace, B. C. (2020). ERASER: A Benchmark to Evaluate Rationalized NLP Models. ACL 2020.

Wang, Y., Mansurov, J., Ivanov, P., et al. (2024). SemEval-2024 Task 8: Multidomain, Multimodel and Multilingual Machine-Generated Text Detection. SemEval-2024.

Models and tools: Llama 3.1 (Grattafiori et al., 2024), Qwen2.5 3B (Yang et al., 2024), Gemini (Google), Nomic Embed (Nussbaum et al., 2024).

Full reference list (33 entries, each verified against the primary source) in the dissertation draft.